

Digital Asset Market Behavior Intelligence Platform

A behavior-explanation platform for crypto markets — regimes, events, sentiment, liquidity, on-chain.

Author: Xu Ao | **Coverage:** BTC, ETH, SOL + 6-token DeFi basket, daily, 2023-01-01 → today **Repo:** github.com/bobaoxu2001/Digital-Asset-Market-Behavior-Intelligence-Platform **Live dashboard:** local — [make dashboard](#) | **Demo mode:** [make demo](#)

Problem

Most crypto research tools either (a) predict next-day price using one or two signals, or (b) display single-source dashboards (price-only, sentiment-only). Neither is what a **digital-asset strategy analyst** actually needs. The analyst's job is to **explain** market behavior, **rank** event sensitivities, **detect** regime shifts, and **produce** strategy-relevant memos — across multiple coupled data sources.

What it is

A reproducible end-to-end platform that fuses six data domains into a single feature table, classifies each (date, asset) day into one of seven interpretable behavior regimes, runs a formal event study over a curated calendar, and surfaces all of it through a polished six-page Streamlit dashboard plus a written research memo.

Data sources (all free-tier; all with documented fallbacks)

Domain	Primary	Fallback
Price / volume / market cap	yfinance + CoinGecko Demo	each fills the other's gaps
Sentiment	Alternative.me Fear & Greed	synthetic series if API down
DeFi liquidity / TVL	DeFiLlama (chains, protocols, stablecoins)	—
BTC on-chain	Blockchain.com Charts	—
ETH on-chain	Etherscan stats	DeFiLlama TVL flux proxy (<i>documented</i>)
Macro	FRED (VIX, DGS10, DXY proxy, SP500)	yfinance equivalents

Domain	Primary	Fallback
Events	Curated CSV — 46 real events, 2023-Q1 → today	—

Methodology

- ~50 daily features per (date, asset): returns, realized vol, volume z-score, drawdown, sentiment z-score & shifts, TVL changes, composite **liquidity stress score**, **on-chain activity index**, macro layer, event flag/severity/days-since-or-to.
- **Rule-based regime classifier** with explicit precedence: Event-driven → Liquidity Stress → Risk-off → On-chain Activity Spike → Momentum → Calm → Neutral.
- **Event study**: per (event, asset) — $t-1 \rightarrow t+1$, $t \rightarrow t+3$, $t \rightarrow t+7$ cumulative log returns; pre/post realized vol & volume; sentiment shift; TVL & on-chain reaction; composite impact score.
- **Lead-lag**: cross-correlation of $\Delta \text{sentiment}_t$ vs $\text{return}_{\{t+k\}}$ for $k \in [-7, +7]$.
- **Look-ahead control**: every rolling feature is trailing-only; regime labels at time t use only features known by end-of-day t .

Key findings

#	Finding	Implication
1	Sentiment lags price by 1 day universally — peak corr at lag = -1 for all 9 assets (BTC +0.65, ETH +0.55, SOL +0.49).	Treat F&G as a <i>confirming</i> indicator, not an alpha.
2	Protocol Upgrades and Exchange events generate the largest impact (mean impact 1.44 and 0.86); ETFs rank lowest (0.22) — already priced in.	Pre-position around tech catalysts; fade hype around scheduled macro / ETF flows.
3	Regime labels separate behavior — <i>Calm</i> days quietly drift negative; <i>Momentum</i> / <i>Event-driven</i> days carry the upside.	Regime label is a useful conditioning variable for any strategy overlay.

#	Finding	Implication
4	Liquidity stress + abnormal on-chain activity coincided with the worst drawdowns (March 2023 SVB / USDC depeg, August 2024 yen carry unwind).	Joint occurrence is a hedge / fade trigger.

Strategy implications

1. **Don't trade Fear & Greed directionally.** Use it inside the regime engine, not as a forward signal.
2. **Pre-position around Protocol Upgrades and Exchange events,** not ETFs or CPI prints.
3. **DeFi tokens (UNI, AAVE, LDO, MKR, CRV) amplify event-driven moves** — higher-beta event expressions when conviction is high.
4. **Liquidity stress + on-chain abnormal activity = fade or hedge.**

Why this matters for digital asset research teams

- **Multi-modal**, not single-source. Replicates how a real analyst combines price, sentiment, liquidity, on-chain, and event data.
- **Interpretable**, not black-box. Every regime label is auditable; every event ranking has a formula a PM can disagree with.
- **Defensible under interview pressure.** Look-ahead bias controlled; limitations documented; fallback paths declared up front.
- **Reproducible end-to-end.** make ingest && make features && make analysis && make dashboard on a fresh clone, in under 2 minutes.

Tech stack

Python 3.11 · pandas · numpy · scipy · plotly · streamlit · pyarrow · yfinance · requests · pyyaml · dotenv · pytest · GitHub Actions CI · DeFiLlama · CoinGecko · FRED · Blockchain.com Charts · Etherscan · GDELT 2.0.

Export to PDF: `pandoc reports/one_page_case_study.md -o reports/one_page_case_study.pdf` (requires pandoc + a LaTeX engine, e.g. tectonic or xelatex).